

Exploring the Feasibility of a Participatory Data Physicalization of CO₂e Emissions for Dietary Choices in Collective Catering Settings

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We present the design and early on-site deployment of a **participatory data physicalization** in which patrons of a collective restaurant report their **dietary choices** and view their associated carbon footprint.

The deployment was held during **one week** in a collective cathering space with a number of patrons varying from **184 to 224** per lunch.



Examples of how the panel was filled during two lunches. On the right, the beef consumption created an overflow.

The design of the panel is inspired from Edo [2] Input Visualization. The design considerations include:

- Empty panel already shows data: CO₂e per protein source
- Accomodate a **large number** of users
- **Short contribution time**
- **Lower complexity** of the visualization
- Allowing **comparison** between meals' CO₂e impact

We did a one-week early deployment in which we observed the following behaviors with patrons:

- **Surprise** - With large values for CO₂e
- **Autonomy** - After some days
- **Engagement** - Increased with a « funny » question
- **Bad pairs' reactions** - with people choosing beef
- **More consideration** - for the vegetarian option

Instructions are displayed above the panle to help consumers to understand what they should do fast.

What did you eat today?

- Use your token to indicate the main dish you had: beef, pork, poultry/fish, or vegetarian.
- The height of each column shows the total CO₂ impact of all meals of that type served today in this restaurant

Sometimes, a question was added to encourage consumers' participation

Daily question: Are you vegetarian?

● YES ● NO

Each pictogram represents one type of protein source and corresponds to one of the above columns.

A scale on the left side indicates the quantity of CO₂ reached for each column

The tokens hold two colors to let users anwsering questions, maximizing their engagement

